

□ EduTracker ERP: Professional Client Deployment Guide

This guide provides a step-by-step procedure for deploying the EduTracker Enterprise ERP. It is divided into two parts: steps to take on your developer machine and steps to take at the client's office.

✂ Phase 0: Developer Preparation (On YOUR PC)

Before going to the client, you must package the application to hide the source code and ensure it's production-ready.

1. **Open PowerShell as Administrator** in the project root.
2. **Run the Release Bundler:**

```
.\bundle_release.ps1
```

- **What this does:**
 - Compiles the Frontend (.next).
 - Compiles the Backend (dist).
 - Copies only the necessary production files to a folder named EduTracker_Release.
 - Excludes all source code (src folders) to protect your intellectual property.
 - 3. **Copy to USB:** Copy the entire EduTracker_Release folder to a USB drive or cloud storage to take to the client.
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🔑 Phase 1: Prerequisites (On Client's PC)

Ensure the client's "Server PC" has the following software installed:

1. **Node.js (LTS):** Download from nodejs.org.
 2. **PostgreSQL (v16+):**
 - Set the postgres password (default recommended: 123456).
 - Create an empty database named edutrackr.
 3. **Static IP:** Recommended for consistent access from other devices in the network.
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□ Phase 2: File Preparation

1. Copy the EduTracker_Release folder (or the project root) to the client's PC (e.g., C:\edutrackr).

2. Open a terminal (PowerShell as Administrator) in that directory.
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Phase 3: Database & Core Setup

We have created a dedicated script to handle the database setup cleanly for clients, installing only the necessary professional templates and a default admin user.

3.1 Run the Database Setup Script

1. Navigate to the backend folder:

```
cd backend
```

2. Run the automated setup tool:

```
.\setup-db-client.bat
```

3. **When prompted**, enter your PostgreSQL URL.

- Example:
postgresql://postgres:123456@localhost:5432/edutracker?
schema=public

4. **What this script does:**

- Generates the database client.
- Creates all required tables (Migrations).
- Installs **15 Professional Document Templates** (ID Cards & Certificates).
- Creates the **Default Admin Account**.

3.2 Default Credentials

After the script finishes, you can log in with:

- **Email:** admin@school.com
 - **Password:** admin123
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⚙️ Phase 3: Environment Configuration (.env)

The .env files control how the app connects to the database and sends emails.

3.1 Backend .env (Critical)

The setup-db-client.bat script creates a basic .env file for you, but you **MUST** edit it to add real email credentials for notifications to work.

1. Open backend/.env in a text editor (like Notepad).
2. Update the following values:
 - SMTP_USER: The school's Gmail address (e.g., schoolname@gmail.com).

- SMTP_PASS: The Gmail **App Password** (not the regular password).
- FRONTEND_URL: If accessing from other PCs, change `localhost` to the Server's IP (e.g., `http://192.168.1.10:6001`).

3.2 Frontend .env (Root Folder)

1. Open `.env` in the root folder.
 2. Ensure `NEXT_PUBLIC_API_URL` is set to `"/api"`. This allows the built-in proxy to handle requests correctly.
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□ Phase 4: Network & Firewall Configuration

To allow other computers in the school to access the ERP:

1. Open **Windows Defender Firewall with Advanced Security**.
 2. Go to **Inbound Rules > New Rule**.
 3. Select **Port > TCP**.
 4. Enter specific ports: `5000`, `6001`.
 5. Select **Allow the connection**.
 6. Name it: `EduTracker ERP Access`.
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□ Phase 5: Launching the System

1. Go back to the project root directory.
2. Run the system start command:

```
.\START_SYSTEM.bat
```

3. Accessing the App:

- **On Server:** `http://localhost:6001`
 - **On Other PCs:** `http://[SERVER_IP]:6001` (e.g., `http://192.168.1.10:6001`)
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✂ Phase 6: Customizing the School Profile

Once logged in as Admin:

1. Go to **Settings > School Profile**.
2. Upload the **School Logo** and **Principal's Signature**.
3. Update the **School Name, Address, and Phone**.

4. These details will automatically appear on all generated **ID Cards** and **Certificates**.
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☐ **Maintenance & Backups**

- **Backups:** Use the built-in Backup tool in the Settings module weekly.
 - **Updates:** Simply replace the `backend/dist` and `src` folders with the latest versions and restart the system.
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Guide version: 2.0 (Updated with Professional Template Seeding)